

Exploring Blind People’s Information Needs for Sightseeing Narration Generated by Vision-Language Models

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Abstract

Blind people often enjoy sightseeing with sighted companions, but they may not always describe the surrounding environment aligned to blind people’s interests. We therefore explore Vision-Language Model (VLM)-generated narration as a supplement to accompanied sightseeing, focusing on how narration should balance concrete scene information with impression-based descriptions of atmosphere. We built a prototype that narrates prerecorded egocentric 360-degree sightseeing videos and conducted a formative online study with seven blind participants across five scenarios. Participants valued narration that reduced the need to ask companions for descriptions and could be tailored through prompt revisions and memory-based repetition control. Preferences varied: some participants wanted concrete objects as cues for interpreting places, whereas others wanted holistic impressions; participants also differed on whether the system should directly state atmospheric interpretations or provide cues for users to interpret themselves.

CCS Concepts

• Human-centered computing → Accessibility.

Keywords

accessibility; prompt engineering; Vision-Language Model; sightseeing

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1 Introduction

Blind people often enjoy sightseeing activities, such as window-shopping [1, 9], with support from sighted companions, including

family members, friends, and sighted guides. We refer to sightseeing in such settings as *accompanied sightseeing*. Although companions can provide valuable support, the information they convey may not always match what blind people want to know about a place. Companions’ descriptions can reflect their own interests and preferences [5, 8, 11], and continuously describing the surrounding environment can be burdensome for sighted companions [13]. Thus, accompanied sightseeing could benefit from assistive systems that describe the surrounding environment in ways that reflect blind people’s individual interests and preferences.

In this work, we explore Vision-Language Model (VLM)-generated narration as a way to support accompanied sightseeing. Our goal is not to support independent navigation, such as obstacle avoidance or route guidance, but to help blind people access surrounding information and atmosphere while moving through places with companions. Sightseeing involves more than knowing the names and locations of nearby objects: people may also want to understand the character, mood, or atmosphere of a place. We therefore distinguish between two types of information. We use *what-is-there* information to refer to concrete, observable elements of a scene, such as objects, signs, products, people, store names, and spatial arrangements. We use *how-it-feels* information to refer to impression-based or interpretive descriptions that convey the atmosphere of a place, such as whether it feels historic, luxurious, or modern.

Prior work has investigated several ways to make visual environments accessible to blind people. Navigation systems have used localization techniques and pre-defined maps to provide concise information about nearby points of interest, such as their names and distances [12]. Other work has examined visual question answering for user-captured images [2], access to visual information in daily tasks [6], and recreational activities such as window-shopping [9] and exploration [10]. More recently, VLM-based systems have enabled richer and more customizable visual descriptions, including descriptions that can be adapted to users’ intents, interests, and desired level of detail [3, 4, 6, 15, 16]. Separately, prior work has explored how impression-based descriptions can help blind people appreciate distant landscapes, such as skylines or mountain views, through sound and speech [7]. However, accompanied sightseeing raises a distinct design question: how should a system balance concrete descriptions of what is visually present with impression-based descriptions of how a place feels? Although prior work has



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examined *what-is-there* and *how-it-feels* descriptions in different contexts, it remains unclear what kinds of information blind people want during accompanied sightseeing, how such information should be conveyed, and whether a system should directly interpret the atmosphere versus provide cues that allow users to form their own interpretations.

To investigate these questions, we developed a VLM-based narration system that generates descriptions from egocentric 360-degree sightseeing videos. The system allows the narration style to be modified through prompting, making it possible to explore how different balances of *what-is-there* and *how-it-feels* information affect blind people’s sightseeing experiences. We conducted an online formative study with seven blind participants. Participants listened to VLM-generated narrations for five recorded sightseeing scenarios while imagining that they were walking through the places with sighted companions. Through iterative prompt revisions based on participants’ feedback, we explored what information they wanted from VLM-generated narration, how they wanted that information to be expressed, and how narration could supplement the accompanied sightseeing.

2 User Study

An IRB-approved online user study was conducted with seven blind participants to investigate the information needs of *what-is-there* and *how-it-feels* descriptions and how they should be conveyed during accompanied sightseeing. All interviews were recorded with participant consent and transcribed later.¹ Participant demographics (age, gender; age of blindness onset) were: P1 (53, female; 36), P2 (22, male; 3), P3 (28, female; 22), P4 (49, male; 44), P5 (32, female; 7), P6 (56, male; 8), and P7 (58, male; 3).

We developed a VLM-based prototype system that provides narrations for accompanied sightseeing from egocentric 360-degree videos for the online user study. During the study, pre-recorded videos were played and narrations were generated on the fly so that prompt could be dynamically adjusted based on the participants’ preference.

The study took 180 minutes, and participants were compensated 60 USD.

2.1 Prototype System

The system assumes that a 360-degree camera is positioned in front of the user. We developed prompts that generate both objective (*what-is-there*) descriptions and impression-based (*how-it-feels*) descriptions from the 270-degree field of view covering the user’s front and both sides. The system sends requests to the VLM every two seconds. To avoid repeatedly describing the same content, it maintains a memory of previously spoken narration and incorporates this memory into subsequent VLM requests. For the user study, we used videos recorded by the research team while walking through several sightseeing locations or selected from a dataset (see Section 2.3.2). Narration was generated on-the-fly from these videos, allowing participants to experience the system while exploring different combinations of prompting strategies and memory mechanisms.

¹All of the communications with participants were done in their native language. In this paper, we describe any translated content in the form of “translated content”.

The memory module maintains a record of the narration spoken during approximately the previous 30 seconds, together with the referenced objects and timestamps. These memory entries are included in subsequent VLM requests as contextual information to discourage redundant descriptions. The memory is also used during post-processing to remove repeated information that may still appear because of overlapping requests or timing differences. The memory module is updated only after the text-to-speech (TTS) engine has actually spoken the corresponding narration, ensuring that only delivered information is remembered. In this prototype, users passively listened to the generated narration, and no interaction with the speech interface was provided. Once narration begins, the system always reads the entire generated utterance without interruption. Any narration generated while speech is in progress is stored as the single candidate for the next utterance. This design prioritizes recently captured visual information while avoiding a backlog of outdated descriptions.

We used OpenAI’s GPT-5.4 Mini (2026-03-17) as the VLM. Each request consisted of a 1920 × 540 panoramic image cropped from the 360-degree video, the prompting instructions, and the current memory entries. The model’s reasoning effort was set to low to minimize response latency. The model was instructed to produce structured JSON output containing fields such as *front*, *right*, *left*, *focus_direction*, *summary*, and *new_information*. The system compared these outputs with the remaining memory entries, merged only previously unspoken information, and constructed the candidate text for the next narration. Speech synthesis was implemented using the OpenAI Realtime Speech API.

2.2 Procedure

We first explained the scope of the study, including our definition of the sightseeing experience, and conducted a short interview to collect participants’ demographics and prior sightseeing experiences. We then explained our system and the concept of prompting, which can control VLM outputs in the desired manner. Participants listened to the generated descriptions while imagining that they were walking through scenes in the pre-recorded videos. After each playback, the experimenter and participants discussed what information participants wanted to receive. We tested the memory function with the first video scenario, then we tested two prompts that focused on either *what-is-there* or *how-it-feels* information with two video scenarios each. The experimenter updated the prompt and generated a new VLM-based description for the same place using the updated prompt. This prompt-update iteration continued until participants were satisfied with the generated output. Finally, we conducted a post-interview to discuss participants’ overall experiences and suggestions for improving the descriptions.

2.3 Conditions

2.3.1 Prompt. We prepared two types of prompts. *What-is-there* prompt instructs the VLM to identify the objects comprehensively, such as store names, displayed products, and their spatial arrangement. *How-it-feels* prompt instructs the VLM to describe the impression of the space and the adjectives associated with each object, such as how light enters the space, the openness of the space, and the materials of floors and walls.

2.3.2 Videos. We selected five scenarios to represent diverse sightseeing contexts. Because window shopping shares similarities with sightseeing, we included both indoor and outdoor shopping environments. As this study was conducted in Japan, we also included a historic Japanese street as a typical tourist destination in Japan. To increase the diversity of the scenarios, we included a historic street outside Japan.

- (1) **Concourse connecting several shopping buildings.** Includes the least change within the video and was used as a case to examine how memory function should handle previously mentioned information.
- (2) **Retail floor in a shopping mall.** Contains dense products, store names, signs, *etc.*, allowing us to ask what object and text information participants wanted and how such information should be described.
- (3) **Modern luxury shopping street.** Contains dense store names, signs, colors, and displays in an outdoor setting. Used to examine what object and text information contribute to the atmosphere of a modern shopping street.
- (4) **Historic shopping street in a domestic context.** Provides a visually distinct outdoor environment and was used to examine how descriptions could convey the atmosphere of a historic setting within the domestic cultural context.
- (5) **Historic street in a non-domestic context.** Used to examine the sightseeing experience in a historic setting outside the domestic cultural context.

The videos for scenarios (1)–(4) were recorded by the research team, and the video for scenario (5) was obtained from the Multi-Modal Social Human Navigation dataset [14].

3 Results and Discussion

3.1 What Makes VLM Narration Useful During Sightseeing?

All participants valued two aspects of VLM-generated narration for accompanied sightseeing. First, the narration could help users obtain surrounding information without having to ask companions for descriptions, allowing them to notice points of interest and use them as topics of conversation. For example, P5 said, “*When I travel with my friends, I want to know about my surroundings, but I feel awkward asking them to describe the surroundings because it would interrupt our conversation. Therefore, I would like to use this system to enjoy both information about my surroundings and conversation with my friends.*” Second, prompt revision and memory function are valued as a way to filter out familiar or irrelevant information and adjust the narration to the place and their current interests by all participants. P1 noted prompt revision’s usefulness, “*The information I want differs depending on the place, so it was useful that the types of information could be adjusted through prompt revisions.*” And P7 noted the value of memory function “*(When memory function is taken on,)It was helpful that it read aloud when the surrounding scenery changed, allowing me to understand changes in my environment.*”

3.2 What Makes Atmosphere Understandable?

Participants differed in whether they prioritized understanding *what-is-there* (P2, P5, P6, and P7) or *how-it-feels* (P1, P3, and P4) when interpreting atmosphere during sightseeing. The *what-is-there* group emphasized concrete elements in the scene, such as buildings, people, and products, as cues for inferring what kind of place they were in. They often redirected default *how-it-feels* prompts toward detailed descriptions of concrete objects of interest. For example, Fig. 1 shows how P2’s request for more detail about “what kinds of products are there” led us to emphasize products in a shopping mall retail-floor scene.

By contrast, participants who prioritized *how-it-feels* wanted to understand the overall impression or image of the place rather than accumulating object-by-object information. These participants often revised prompts toward broader, more global descriptions. For example, Fig. 1 also shows a contrasting revision in the same scene: P3’s request for “the overall impression” led us to emphasize scene-level atmosphere rather than concrete objects.

These differences appeared to align with participants’ visual experience. Participants with early-onset blindness (P2, P5–P7) were generally characterized by the *what-is-there* approach; for example, P2 rarely inferred atmosphere from visual cues such as design, so knowing what was present contributed more directly to his enjoyment. In contrast, participants with later-acquired visual impairment (P1, P3, P4) tended to seek holistic impressions, such as by connecting descriptions with visual memories. This suggests that future atmosphere-description systems may need to support different interpretive strategies depending on users’ visual experience, although further research is needed to examine whether this pattern generalizes beyond our small sample.

3.3 Who Should Interpret How It Feels?

Participants differed in whether the VLM should directly interpret and express how a place feels or provide cues that allow users to interpret the atmosphere themselves. Three participants (P5–P7) valued direct *how-it-feels* descriptions, especially when the VLM made the basis of its interpretation clear. For example, P5 valued descriptions that linked impression words such as “solemn” or “gorgeous” to specific stores or objects, because this helped them understand how visible elements contributed to the overall atmosphere. In contrast, three other participants (P2–P4) preferred the VLM to provide cues rather than directly interpret the atmosphere. P2 prioritized concrete information about *what-is-there*, while P3 and P4 preferred to form their own impressions from the VLM’s descriptions rather than receive direct *how-it-feels* interpretations. For example, P4 said, “*When I think back to when I could see, I experienced the atmosphere of a place by looking at it together with the emotions it evoked.*”

4 Limitation

We were unable to prepare and use videos covering all possible sightseeing environments. For example, the videos did not include rural or nature-rich environments. In addition, because the study was conducted online in one-on-one sessions with blind participants, it did not examine how AI narration affects interactions between blind people and their companions while walking together

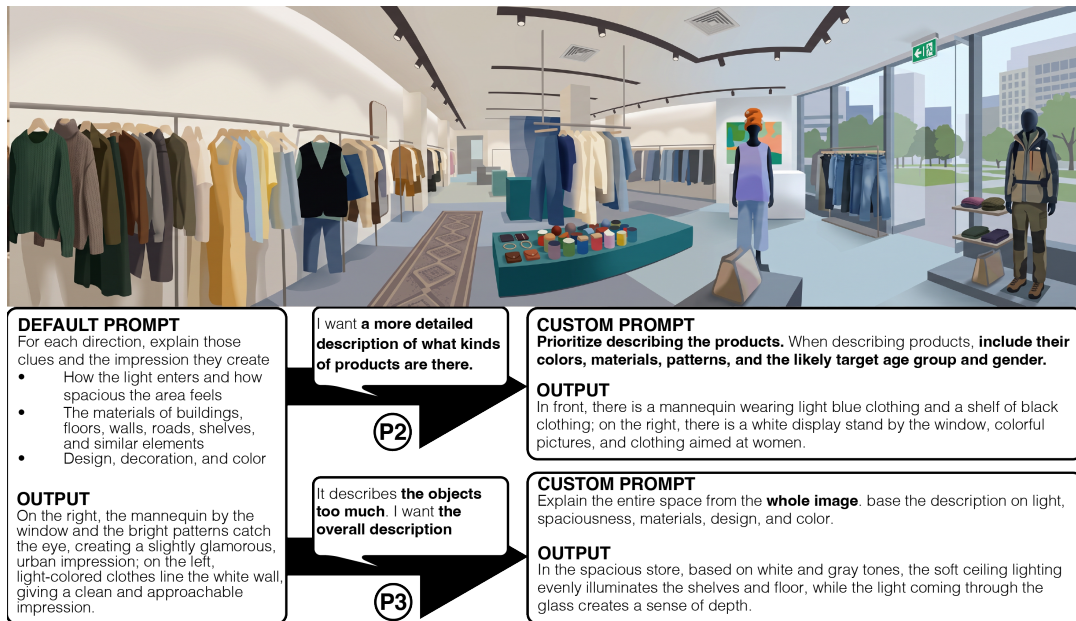


Figure 1: The process of incorporating user requests into prompts.

at actual sightseeing sites. Investigating this effect remains future work. Finally, the observed tendencies in visual memory and the understanding of atmosphere should be examined with a larger sample.

5 Conclusion

We investigated blind people's information needs for VLM-generated narration during accompanied sightseeing. Participants valued narration that could both provide surrounding information without requiring users to ask companions for descriptions and be tailored through prompt revisions. Their preferences varied in the balance between *what-is-there* and *how-it-feels* information and in whether the VLM should interpret atmosphere directly or support users' own interpretation. These preferences appeared to be related to participants' visual experience. Future work will evaluate an on-site system and examine how visual experience shapes these needs.

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